

The Value of Off-Exchange Data

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Off-exchange data is valuable

Public dissemination of off-exchange trades triggers quote and trade activities

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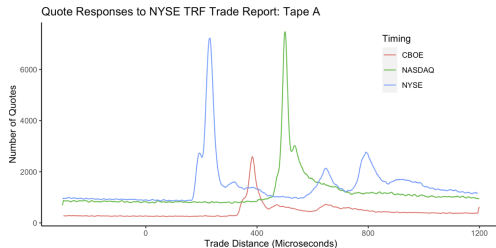
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Identification: Are these responses indeed triggered by off-exchange data?

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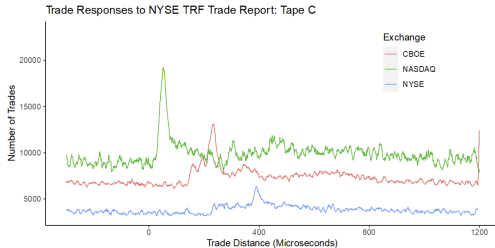
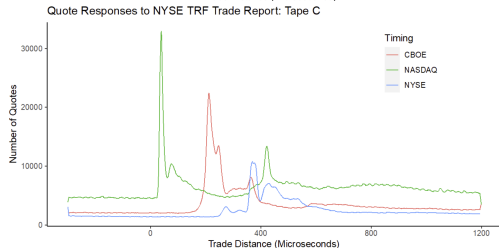
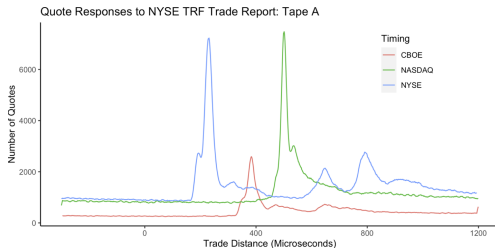
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Percentage Trades with Same-Direction

	1 st Quartile	Mean	3 rd Quartile
Trades	64%	65%	67%
Quotes	59%	61%	64%

Suggestion #1: Implications for measuring the relative informativeness of off-exchange trades?

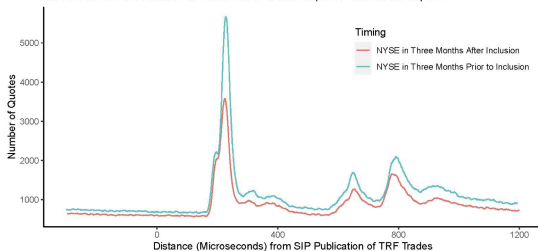
“Upstair trades” are less informed than on-exchange trades:

Empirics Rose (2014), Madhavan and Cheng (1997), Booth, Lin, Martikainen and Tse (2002), Bessembinder and Venkataraman (2004), Bernhardt, Dvoracek, Hughson and Werner (2005), Smith, Turnbull and White (2001), Westerholm (2009)

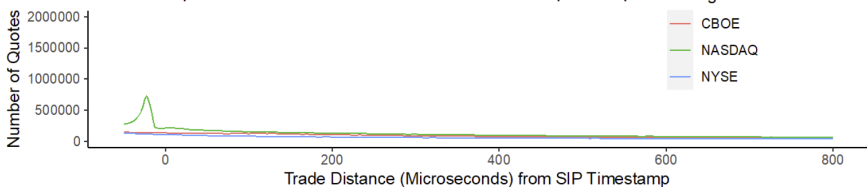
Theory Seppi (1990), Foucault and Desgranges (2005), Zhu (2014), Lee and Wang (2017)

Observation #1: Why traders do not read the NYSE/Nasdaq proprietary data feeds for off-exchange trades?

Quotes on NYSE Relative to NYSE TRF Trade Report Publication: Tape A



Quote Responses SIP Pub. of NASDAQ EX Trade Report: Tape A



Observation #2: How does the value of off-exchange data survive long latency?

The parties to the original off-exchange trade (say the dealer) are immediately notified of the trade.

Why do they not take advantage and drive the value of data to zero?

They could also snipe stale quotes on exchanges.

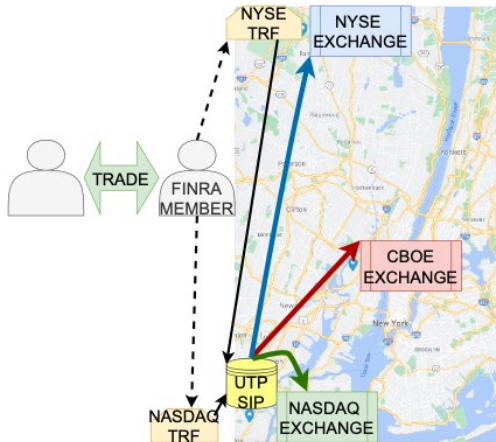
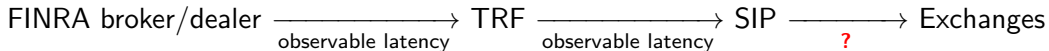
Suggestion #2: Value of data is concentrated on certain types of off-exchange trades and takes time to exploit?

- Variation of response/realized spread across different types of off-exchange trades
 - ▶ large vs. small
 - ▶ institutional vs. retail
 - ▶ broker-internalized (Robinhood/Drivewealth, Bartlett, McCrary, and O'Hara, 2022) vs. dealer-internalized (Citadel/Virtu/Two-sigma)
 - ▶ long vs. short latency from broker/dealer to TRF.

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- Measure how quickly the value of data decays
 - ▶ Consider the “half-life” of the spike in responses
 - ▶ Is the half-life long than the latency?
 - ▶ Variation in the half-life across different types of off-exchange trades

Suggestion #3: Possible to obtain latencies from SIP to exchanges?



Summary

- Novel and well-identified patterns showing that
 - ▶ off-exchange data is valuable
 - ▶ the value survives long latencies in trade reporting and dissemination
- Thought provoking about how the value of off-exchange data survives long latency?
- Main suggestion: Analyze how the value of data
 - ▶ varies across different types of off-exchange trades
 - ▶ decays over time